

45-Day PHP Logical Coding Challenge - Full Checklist

Week 1: Basics & Logic Foundation

- Day 1: Print all even numbers from 1 to N (without using built-in filter functions)
- Day 2: Print all numbers divisible by 3 and 5 in range 1 to 100
- Day 3: Implement Linear Search manually
- Day 4: Find the second largest element in an array (no hints)
- Day 5: Find the second largest element in an array with duplicates (no built-in sort)
- Day 6: Count even and odd elements in an array (no built-in filter)
- Day 7: Count frequency of each element in an array (use associative arrays, no built-in frequency functions)

Week 2: Intermediate Logic + Patterns

- Day 8: Reverse an array without using built-in reverse function
- Day 9: Check if a number is a Palindrome
- Day 10: Check if a number is an Armstrong number
- Day 11: Find the sum of digits of a number
- Day 12: Find factorial of a number (without recursion)
- Day 13: Find greatest common divisor (GCD) of two numbers
- Day 14: Swap two numbers without using a third variable

Week 3: String and Pattern Handling

- Day 15: Count vowels and consonants in a string
- Day 16: Reverse a string without using `strrev()`
- Day 17: Check if a string is a Palindrome
- Day 18: Count frequency of characters in a string
- Day 19: Print triangle pattern using stars (*)
- Day 20: Print inverted triangle pattern using stars
- Day 21: Print half pyramid using numbers

Week 4: Array & Number Series

- Day 22: Merge two arrays without built-in merge functions
- Day 23: Sort array elements manually (no built-in sort)
- Day 24: Remove duplicate elements from array
- Day 25: Check if a number exists in an array (without `in_array()`)
- Day 26: Rotate array to the right by K steps

- Day 27: Rotate array to the left by K steps
- Day 28: Find intersection of two arrays (manually)

Week 5: Advanced Number & Logic Series

- Day 29: Check whether a number is prime or not
- Day 30: Print first N prime numbers
- Day 31: Generate Fibonacci series up to N terms
- Day 32: Check if a number is part of Fibonacci series
- Day 33: Find LCM of two numbers
- Day 34: Find the sum of array elements (no built-in array_sum)
- Day 35: Check if two strings are anagrams

Week 6: Practice Mix + Real Logic

- Day 36: Find all pairs in array whose sum is equal to target
- Day 37: Move all zeros to the end of the array
- Day 38: Find missing number in array 1 to N
- Day 39: Count number of digits in an integer
- Day 40: Check if number is a perfect square (without sqrt())
- Day 41: Create multiplication table for any number
- Day 42: Sum of even and odd digits of a number

Week 7: Final Challenge Days

- Day 43: Separate positive and negative numbers in array
- Day 44: Print unique elements from array
- Day 45: Final Day Project - Mini Student Marks Manager (Array-based, calculate average, max, min, grade logic)